

DASE7506 Project 1 - Stage143 final report

1. Abstract

We train a small causal language model from random initialization using only the supplied WikiText-2 training text and fixed 2,048-token BPE tokenizer. The frozen model combines an eight-block alternating-attention/causal-convolution Transformer, within-window prefix copy, and train-only order-six modified Kneser-Ney (MKN) statistics through a causal gate. OpenVINO evaluates its frozen neural features on CPU while the compact PyTorch head, copy mechanism, count tables and gate produce one normalized 2,048-way distribution per position. The latest Windows CPU FP32 *validation* score is **1.399686162 BPB** over 376,599 next-token targets. Three fresh-process local CPU comparisons yield **3.617702x** baseline time, **2,176,729,088 bytes** maximum peak RSS and **55,810,412 bytes** of conservatively counted inference assets. Frozen Stage143 complete-test CPU FP32 BPB: **1.415657617** over **428,405** targets and **1,292,013** raw UTF-8 bytes. An independent Linux one-thread host exceeded the 5x CPU-time limit; this Windows qualification is not a cross-platform guarantee.

2. Task, data and evaluation

The [assignment guide](#) ranks complete-test bits per raw UTF-8 byte under a fixed scorer, independent 256-token causal windows, CPU FP32 scoring, at most 5x baseline CPU time, at most 4 GiB peak RAM and at most 64 MiB of uncompressed inference assets. Training may use CPU/GPU, but only the supplied training text; validation is for development and test must wait until the method is frozen. No external text or pretrained weights were used in the experiments reported here. The model does not retain state across evaluation windows or call a network during inference.

The original course baseline measured **2.102014912 full-test BPB** in the earlier frozen comparison. A previously frozen Stage10 model produced **1.605136467 full-test BPB**, but that is a different checkpoint and must not be presented as Stage143's test score. The frozen Stage143 result above improves on the original baseline by **32.65%** in BPB on the same complete test split; the development validation score is reported separately.

3. Model and training mechanism

The neural ancestor has width 288, eight blocks, eight attention heads and context 256. Blocks 2/4/6/8 use kernel-seven gated causal depthwise convolution, alternating with global causal self-attention. The model uses RoPE, RMSNorm, SwiGLU, tied token/output embeddings and a within-window prefix-copy distribution. The fixed [Stage54 configuration](#) records dropout 0.1, embedding-row dropout 0.1, deep-supervision layers 4/6, future-token offsets 2/3 and R-Drop coefficient 0.5. The key output statistics are modified Kneser-Ney tables derived from the supplied train text; a four-feature gate blends them with the neural/copy distribution. The gate's learned statistics are train-derived, while its final feature mask/scale was selected on validation and is disclosed as such.

This is a multi-stage lineage, not one 7,200-step run: Stage54 trains the hybrid neural model; Stage56 continues its fixed average; later output-bias, count-aware and distillation stages change the neural expert and gate before the exact Stage105 predictor is packaged as Stage143. All checkpoints in the lineage inherit their prior training cost. The detailed schedules, seeds, target presentations and negative screens are retained in `code/docs/`, `code/results/` and the code scripts. A hash-linked [accepted-lineage audit](#) totals **255,225,110 documented primary training-target presentations** and **7,754.94 seconds of recorded training time** across Stages54/56/61/63/ 65/67/71/92. R-Drop uses two stochastic passes for many of these targets; the count is not FLOPs or unique corpus positions. This is a **lower bound on the accepted neural lineage**, excluding train-only count construction, rejected/failed searches, validation, export and runtime audits. A complete project-wide hardware-time bill remains unavailable. A separate [archive audit](#) enumerates 65 distinct metrics files: 63 comparable `train_seconds` fields sum to **53,553.04 seconds (14.88 hours)** of reported training, including the accepted lineage and rejected trials. The Stage136 record reports only broader elapsed time; the Stage169 aggregate record has no `train_seconds` because its two arm-specific metrics records supply those durations. Both aggregate-only records are excluded from the sum. The sum omits unarchived/failed runs, validation, preflights, table construction and system overhead, so it must not be mistaken for a complete hardware bill.

Stage143 changes only inference packaging relative to Stage105: it stores one 31,805,041-byte ONNX feature graph and a 23,824,895-byte checkpoint containing the head, copy, MKN and gate tensors, avoiding duplicate backbone weights. There is no new training, coefficient fitting or validation selection in the OpenVINO export. The model still returns finite, normalized causal log-probabilities for the unchanged scorer.

4. Equal-target controls and ablations

The following completed runs use seed 17, batch 32, **7,200 updates** and **58,982,400 primary next-token presentations**; the same sampled-window sequence, optimizer settings and five-checkpoint averaging rule make the paired comparisons interpretable. The optimization trajectories differ by design. R-Drop processes each primary position twice stochastically; this extra training compute is not a new independent target count.

Model	Complete CPU FP32 validation BPB	Controlled question
Stage26, ordinary objective	1.464993908	Reference for training-only R-Drop
Stage47, same architecture + R-Drop	1.450613003	R-Drop gain 0.014381 BPB, no inference change
Stage54, R-Drop + alternating conv/attention, width 288	1.428594045	Matched Stage47 comparison; gain 0.022019 BPB

Stage47 versus Stage26 isolates the training objective; Stage54 versus Stage47 tests the **combined** backbone reallocation (convolution placement and width/FFN capacity), not the effect of convolution alone. These controls do not share the later continuation/distillation budgets of Stage143 and must not be presented as an equal-budget Stage143-versus-baseline comparison. Source records: [Stage47](#), [Stage54](#).

The later, separately budgeted path improves the development predictor: Stage54 neural-only 1.428594045; Stage56 continued neural 1.420423862; Stage71 neural+MKN 1.406959541; Stage85 resource-qualified calibrated mixture 1.403024133; the Stage103/105 gated model 1.399686163; Stage143 packaging 1.399686162. These numbers are **validation model-selection evidence**, not matched-target ablations or independent replications of the final score. The Stage143 score still misses the aspirational 1.35 target.

5. Resource and reproducibility audit

The exact Stage143 checkpoint passed the unchanged evaluator on full Windows CPU FP32 validation. In three alternating fresh-process comparisons, baseline times were 23.7861/23.8256/23.3136 s and candidate times were 86.8676/86.0511/85.9281 s. The ratio of medians was 3.617702x. The maximum candidate peak working set was 2,176,729,088 bytes; conservative assets were 55,810,412 bytes. These pass the three limits on the measured Windows host, but do not guarantee the instructor's CPU timing. The [Stage143 qualification record](#) contains exact hashes and individual measurements. A fresh directory extraction reproduced 1.399686162 BPB with the tracked assets. An independent [Linux x86-64 CI run](#) reproduced **1.399686179 BPB** over the same complete validation split, using one actual FP32 OpenVINO thread. A separate three-repeat, same-host Linux one-thread resource check measured **5.502555x** baseline CPU time, above 5x; RAM and assets passed. A four-thread recheck was unavailable on that two-logical-CPU host because OpenVINO reported one inference thread, so neither a four-thread pass nor failure was measured. An unchanged-graph ONNX Runtime pilot improved sampled feature time only 5.31%, below its 12% gate. The Windows pass therefore is not universal CPU eligibility; see the [Stage172](#) and [Stage173](#) records. A Stage187 node profile found dense FP32 projections dominated sampled feature time; no qualifying local rewrite followed. The separate [pre-test readiness audit](#) rechecked the frozen-file *candidate* hashes, score coverage and Windows resource arithmetic. It is a consistency check, not method freeze, test evaluation or final submission.

The frozen checkpoint and feature graph are pinned separately by the committed pre-test freeze record:

code/checkpoints/stage143-openvino-order6.pt (23,824,895 bytes):

```
256e0e3cd32e23c8ef2551a80ab3b39fd094401c9034853849da35b9963c6da3
```

code/inference_assets/stage143-stage92-features.onnx (31,805,041 bytes):

```
5da1de435c86b40c97718e8a5bbc990af9f2eab431cb8a907b36fbc3bcce5ef4
```

Reproduction from code/, in an environment installed per `code/README.md`:

```
python scripts/verify_fixed_files.py
python -m unittest discover -s tests -v
python evaluate.py \
  --checkpoint checkpoints/stage143-openvino-order6.pt \
  --device cpu --precision fp32 --threads 4 \
  --split validation --output results/stage143-reproduction.json
```

The method was frozen at committed source revision 3a7befc0aa603745b5e94d317ef9dcdf61f94764 before the SHA-bound `code/frozen-test entry point` produced the complete CPU FP32 score reported above. The freeze record, test JSON and window-loss sidecar bind this result to the exact checkpoint and source hashes. Public code/checkpoint links and peer reproduction require a separate final release audit.

6. Negative results, limitations and AI disclosure

Stage145/147/150 changed attention allocation, width and depth but missed their continuation gates. Stage155's larger neural model reached 1.409877 validation BPB; Stages157/158 distilled a stronger, over-budget teacher only to about 1.403. Spelling residuals, frequency-weighted loss, train-only nearest-neighbor retrieval, dilated convolution and input masking failed their gates (Stages160/161/163-165). Stage168 sharpness-aware training regressed versus its matched Stage54 control (1.522015 vs 1.519950 BPB). Stage169 teacher transfer improved Stage143 by only 0.000246 BPB, below its 0.005 gate; Stage174's byte auxiliary loss was slightly worse than control. None was promoted or test-scored; protocols are in the records.

Stage175's **answer-aware, non-deployable** neural/MKN oracle reached 1.300186476 validation BPB. A legal gate would need about half this hindsight gain to reach 1.35. Stage176's richer causal-feature gate, fitted on one validation half and evaluated on the other in both directions, instead scored 1.400044877 BPB; its validation-fitted coefficients were not deployed. Stage177's parallel attention/convolution Transformer failed its CPU-feature speed gate at 1.438569x the Stage143 graph before training. Stage178 reviewed prior negative output-head comparisons. Stage179's fixed 7,200-step dropout-0.20 control scored 1.444744045 versus the matched Stage54's 1.428594045 BPB, so it stopped. These are not submission scores; Stage180's hidden-matrix Muon optimizer pilot scored 1.534018833 at the matched 2,400-step endpoint, versus AdamW's 1.519950369; its 0.020-BPB advancement gate failed and it cost 1.52x as much training time. Its [adapted algorithm and license](#) are disclosed. Stage181's zero-training intermediate-layer readout gained only **0.0041 bit per target** on a 32,768-target validation prefix, below its 0.02-bit gate. Stage182's matched four-head pilot gained **0.001139 BPB**, below its 0.015 gate. Stage183's normalized geometric expert fusion gained only **0.001439 BPB** on complete validation, below its 0.015 gate; it was not deployed. Stage184's training-only 0.10 block drop path scored **1.537355105 BPB** at its fixed 2,400-step endpoint, worse than its matched Stage54 control **1.519950369 BPB**; it was stopped without a long run. Stage186's four narrow parallel attention paths passed an input-only resource screen but improved the matched 2,400-step endpoint by only **0.005196 BPB**, below its 0.020 gate. These are not submission scores; Stage188's same-support Witten-Bell count swap regressed to **1.414884002 BPB** on complete CPU FP32 validation and was rejected. Stage189/190's fixed-tokenizer support screens improved by only **0.000590/0.000581 BPB**, below their 0.015 gate, and were not deployed. Stage162's sparse two-model oracle is also non-deployable.

The validation score reflects many sequential decisions on one development split and may be optimistic. The course CPU may differ from the Windows laptop. The resource audit excludes third-party runtime wheel sizes but counts every student-authored inference asset. A post-freeze complete-test result is recorded above, but ranking and eligibility on the course CPU remain subject to independent verification. A separate [macOS ARM64 runtime check](#) aborted during OpenVINO CPU-device discovery before model compilation or scoring, so macOS compatibility remains unverified. A student must inspect and understand the code and all claims.

OpenAI Codex provided substantive assistance with assignment analysis, model/experiment design, implementation, testing, Windows execution, diagnostics, documentation and this report. Earlier conceptual work reused RoPE, SwiGLU, RMSNorm, R-Drop, modified Kneser-Ney, prefix-copy and kNN-LM ideas. The implemented concepts' original publications are listed below; citing an idea does not imply that the paper's implementation or pretrained weights were copied. The [detailed disclosure](#) and experiment records distinguish reused ideas, AI contributions and original project-specific implementations. Passing automated checks is not a substitute for student review.

7. Method references

- Su et al. (2021), [RoFormer: Enhanced Transformer with Rotary Position Embedding](#) - RoPE.
- Shazeer (2020), [GLU Variants Improve Transformer](#) - SwiGLU.
- Zhang and Sennrich (2019), [Root Mean Square Layer Normalization](#) - RMSNorm.
- Liang et al. (2021), [R-Drop: Regularized Dropout for Neural Networks](#) - stochastic consistency objective.
- Chen and Goodman (1999), [An Empirical Study of Smoothing Techniques for Language Modeling](#) - modified Kneser-Ney smoothing.
- Merity et al. (2016), [Pointer Sentinel Mixture Models](#) - within-context copy concept.

The later compact kNN-LM pilot was rejected and is documented in the [Stage163 record](#), which cites its originating publication. It is not part of the Stage143 predictor.